

Exercise for Sect. 2.6

Exercise

2.6.1

Let $X = \{x_1, \dots, x_r\}$ be a finite set, and let $\sigma : X \rightarrow X$ be a permutation of X . The orbit of x_j under σ is the set $\{\sigma^n(x_j)\}_{n \geq 0}$, and σ is called cyclic if there is an orbit of cardinality r .

1. For a cyclic permutation σ and any function $f : X \rightarrow \mathbb{R}$, prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{r} (f(x_1) + \dots + f(x_r)).$$

2. More generally, prove that for any permutation σ and function $f : X \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{p_x} (f(x) + f(\sigma(x)) + \dots + f(\sigma^{p_x-1}(x)))$$

where the orbit of x has cardinality p_x under σ .

Proof: Let $(X, \mathcal{P}(X), \mu, \sigma)$ be a measure-preserving system by defining $\mu(B) = \frac{1}{r} |B|$ for any set in $\mathcal{P}(X)$. σ is ergodic if it is cyclic.

1. By the ergodic theorem, for any $f : X \rightarrow \mathbb{R}$ (it automatically satisfies measurable and L^1), we have $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f \circ \sigma^j(x) = \int f(x) d\mu = \frac{1}{r} (f(x_1) + \dots + f(x_r))$.
2. Just restrict the space to the orbit of x .

□

Exercise

2.6.2

Minic the proof of finite Vitali covering lemma to prove that for any collection of intervals

$$I_1 = [a_1, a_1 + l(1) - 1], \dots, I_K = [a_K, a_K + l(K) - 1]$$

in $\mathbb{Z}$ there is a disjoint subcollection $I_{j(1)}, \dots, I_{j(k)}$ such that

$$I_1 \cup \dots \cup I_K \subseteq \bigcup_{m=1}^k [a_{j(m)} - l(j(m)), a_{j(m)} + 2l(j(m)) - 1].$$

Proof: Let $R = \{1, 2, \dots, K\}$ be the remaining set. Let $j(1) \in R$ that $l(j(1))$ is the biggest one in $\{l(k) : k \in R\}$. Then, remove the elements e in R such that $I_e \cap I_{j(1)} \neq \emptyset$. We have $I_e \subseteq [a_{j(1)} - l(j(1)), a_{j(1)} + 2l(j(1)) - 1]$ since $l(e) \leq l(j(1))$ and $|a_e - a_{j(1)}| < l(e) + l(j(1)) \leq 2l(j(1))$ by $I_e \cap I_{j(1)} \neq \emptyset$. Then, do the same procedure again and again until there is nothing left in R . We have the disjoint subcollection $I_{j_1}, \dots, I_{j_k}$. $\square$

Exercise

2.6.3

Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system. Prove that, for any $f \in L^1_\mu$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^{-n} x)$$

almost everywhere.

Proof: By the ergodic theorem, we have $f^*, g^* \in L^1$ such that for almost every $x \in X$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = f^*(x), \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^{-n} x) = g^*(x).$$

Also we have $\int f d\mu = \int f^* d\mu = \int g^* d\mu$, f^* is T -invariant and g^* is T^{-1} -invariant.

Besides, we have $\int f^* - g^* d\mu = 0$ and we want to show that $\int |f^* - g^*| d\mu = 0$.

First, we want to show T -invariance is equivalent to T^{-1} -invariance. Suppose A is a T -invariant set, $T^{-1}A = A$. This implies that $TA = A$ and $TA^c = A^c$. Thus, $(T^{-1})^{-1}A = TA = A$ and A is a T^{-1} -invariant set. For a T -invariant L^1 function f , $f \circ T = f$ almost everywhere. Since T is invertible and measure-preserving, compose T^{-1} on both side and get $f = f \circ T^{-1}$ almost everywhere. Therefore, T -invariant function is T^{-1} -invariant.

Next, for any T -invariant set A , restrict the space to A , and use the ergodic theorem again. Then we have $\int_X \chi_A \cdot (f^* - g^*) d\mu = \mu(A) \int_A f^* - g^* d\mu_A = 0$ for any T -invariant set A . (Note that the $\mu_A(B) = \frac{\mu(B)}{\mu(A)}$ is not defined when $\mu(A) = 0$, however, the integral is still zero by the integral on X .)

Then, let $A = \{x \in X : f^*(x) > g^*(x)\}$ and $B = \{x \in X : f^*(x) < g^*(x)\}$. A, B are T -invariant by the invariance of f^* and g^* . By the argument above,

$$\int |f^* - g^*| d\mu = \int_A f^* - g^* d\mu - \int_B f^* - g^* d\mu = 0 + 0 = 0.$$

Thus, $\|f^* - g^*\|_1 = 0$ implies $f^* = g^*$ in L^1 and $f^* = g^*$ almost everywhere. $\square$

Exercise

2.6.4

Fill in the details to prove the estimate in (2.28).

Proof: Clearly $A_N(F_0) = F_0$ since F_0 is T -invariant, while if M is fixed and $N \rightarrow \infty$ we have

$$\begin{aligned} \|A_N(A_M(f_0)) - A_N(f_0)\|_\infty &= \left\| \frac{1}{NM} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} f_0 \circ T^{n+m} - \frac{1}{NM} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} M \cdot f_0 \circ T^n \right\|_\infty \\ &= \frac{1}{NM} \left\| \sum_{k=0}^{N+M-2} (c_k - d_k) f_0 \circ T^k \right\|_\infty, \end{aligned}$$

where

$$c_k = \begin{cases} k+1 & \text{if } 0 \leq k < M-1 \\ M & \text{if } M-1 \leq k \leq N-1, \\ N+M-1-k & \text{if } N \leq k \leq N+M-2 \end{cases}, \quad d_k = \begin{cases} M & \text{if } 0 \leq k \leq N-1 \\ 0 & \text{if } N-1 < k \end{cases}.$$

Thus,

$$\begin{aligned} \|A_N(A_M(f_0)) - A_N(f_0)\|_\infty &= \left\| \frac{1}{NM} \sum_{k=0}^{M-2} (k+1-M) f_0 \circ T^k + \sum_{k=N}^{N+M-2} (N+M-1-k) f_0 \circ T^k \right\|_\infty \\ &\leq \frac{\|f_0\|_\infty}{NM} \left(\sum_{k=0}^{M-2} (k+1-M) + \sum_{k=N}^{N+M-2} (N+M-1-k) \right) \\ &= \frac{\|f_0\|_\infty}{NM} \left(\frac{M(M-1)}{2} + \frac{M(M-1)}{2} \right) \\ &= (M-1) \cdot \left(\frac{\|f_0\|_\infty}{N} \right). \end{aligned}$$

□

Exercise

2.6.5

Formulate and prove a pointwise ergodic theorem for a measurable function $f \geq 0$ with $\int f \, d\mu = \infty$, under the assumption of ergodicity.

Proof: Suppose that $(X, \mathcal{B}, \mu, T)$ be an ergodic system. Suppose $f : X \rightarrow \mathbb{R}$ be a positive measurable function with $\int f \, d\mu = \infty$. Then, for almost every $x \in X$,
$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \infty.$$

For each $M \in \mathbb{N}$, let $f_M(x) = \min\{f(x), M\}$ be truncated function of f . Since f_M is bounded, $f_M \in L^1$ so use Birkhoff ergodic theorem,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f_M(T^n x) = \int f_M \, d\mu \text{ for } x \in A_M,$$

where A_M is a full measure set. By the definition of truncation, we have for all $x \in A_M$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \geq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f_M(T^n x) = \int f_M \, d\mu.$$

Let $A = \bigcap_{M \in \mathbb{N}} A_M$, A is full measure and

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \geq \lim_{M \rightarrow \infty} \int f_M \, d\mu.$$

By monotone converge theorem, $f_M \rightarrow f$ implies the rhs is infinity. □